

Tobi Obadiah

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PROFESSIONAL SUMMARY

AI/ML Engineer with 6+ years of experience building machine learning systems, backend infrastructure, and distributed cloud platforms. MSc in Artificial Intelligence with hands-on experience in RAG, LLMs, vector and graph databases, and scalable ML pipelines. Proven track record of designing production-grade systems and leading engineering teams across AI, fintech, and blockchain products. Interested in cognitive architectures, agentic AI, and reinforcement learning.

SKILLS

Expert: Python, C++, C, PyTorch, RAG, AWS, PostgreSQL, MySQL, Linux, Graph Databases, Git, OCaml, REST APIs

Professional: FastAPI, Django, Keras, LLMs, LangChain, Vector Databases, Deep Learning, Pandas, Docker, Kafka, Microservices, TypeScript, Node.js, CI/CD, Blockchain

EXPERIENCE

Machine Learning Engineer

11/2020 – 06/2021

Mboa Digital

- Developed an ETL machine learning pipeline for automated bank statement and invoice reconciliation, achieving 90% accuracy across hundreds of documents processed for a financial compliance, payroll, and tax management suite.
- Evaluated XGBoost, Random Forest, Linear Regression, SVMs, and neural networks for financial document matching.
- Integrated Google Document AI and deployed scalable inference pipelines on AWS Lambda and API Gateway.

Backend Engineer

03/2024 – 10/2024

Neptune Technologies LLC

- Built FastAPI services on PostgreSQL for a asset-management CRM used to manage client portfolios.
- Integrated Stripe Subscriptions (checkout, webhooks, dunning/retries) so the CRM collected recurring payments automatically and cut manual invoice handling and failed-charge follow-up.

Full-Stack Developer

04/2022 – 05/2024

N01ZET

- Led engineering team at N01ZET (payments, blockchain tokens), improving performance by 60% with TypeScript and Node.js.
- Decoupled contract, deployment, and verification workflows with Kafka microservices to remove write and verify path latency.

Software and AI Engineer

11/2019 – 10/2020

Artificials & Engineering

- Engineered and tested core features of Ride & Smile, an Uber like transport application using C++, G-Test, Boost and MySQL.
- Built a Python/PySide (Qt) app to visualize and export field reports, replacing spreadsheets for faster machine-learning analysis.

Freelance Software Developer

04/2018 – 05/2020

Upwork / Fiverr

- Delivered software projects for clients across six continents, earning 30+ five-star reviews and 100% client satisfaction.

PROJECTS

Graph Memory for Long-Term Knowledge Retrieval (MSc Dissertation)

- Researched, designed and evaluated a SALM-inspired word-token graph memory in a custom C graph database as LLM RAG.
- Built and benchmarked a matched MiniLM + FAISS vector RAG on multi-hop QA HotpotQA, 2WikiMultihopQA, and SciFact.
- Graph retrieval beat vector baseline +18% MRR, -45% query latency, -85% ingest latency on multi-hop QA; vector led on SciFact.

Graph Database Engine

- Built a high-performance graph database in C with ACID transactions, WAL persistence, bitmap indexes, and vector search.
- Exposes LTM mode for agentic RAG, continuously evolving as an experimental platform for AI long-term memory research.

EDUCATION

De Montfort University

Master of Science in Artificial Intelligence.

09/2026

University of Buea

Bachelor of Science in Computer Science.

03/2019

CERTIFICATIONS

Deep Learning Nanodegree, Udacity

2019

Deep Reinforcement Learning for Enterprise, Udacity

2019